CURRICULUM VITAE

Saheed Faremi

Data scientist · production ML & LLM systems, owned end to end

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Data scientist and founding engineer with production systems in fintech, legal, and agriculture. Built the fraud/AML intelligence layer for a payments platform operating in eight African markets; shipped LLM document automation that cut manual work by 85%; published deep-learning research in Brain Informatics and IEEE. Core stack: Python, SQL, PyTorch, scikit-learn, Azure ML / AI Studio, AWS, Go, TypeScript. PhD researcher in deep learning for EEG at University College Cork. UNESCO India-Africa Hackathon 2022 gold medalist.

EXPERIENCE

Founding engineer · Curnance

Present

Founding engineer at a multi-asset fintech for African markets. Set up the engineering organisation and shipped seven production services (ledger, auth, KYC/KYB, Go jobs handler, Flutter app, admin console) at 1,000+ daily transactions and 99.9% reliability.

- Founded the engineering side and shipped seven production services: wallet/ledger, auth, tiered KYC/KYB, a Go jobs handler, a Flutter mobile app (~120 screens), a SvelteKit admin console, and a Next.js site
- Built a double-entry general ledger that refuses unbalanced entries, scaling to 1,000+ daily transactions at 99.9% reliability across eight African markets
- Rebuilt all external payout paths debit-before-pay with row-level locking and deterministic idempotency keys after a live withdrawal-fraud incident
- Designed a rules-first, LLM-last fraud/AML intelligence layer: 38 deterministic detectors, LLM-drafted suspicious-activity reports, and a compliance Q&A analyst guarded by k-anonymity floors and PII redaction
- Integrated three payment providers (VFD, Flutterwave, Paystack) with fail-closed webhook verification and per-provider kill switches
- Automated build and deployment with per-service Azure DevOps pipelines and Bicep infrastructure, reducing time to production by 40%; quality gates include 680+ automated tests and formal VAPT remediation
- Now building an in-app assistant that maps natural-language banking requests (balance, exchange rate, recent transactions) to app actions

Data Scientist · Etihuku

2021-05 → present

Data science and software engineering at Etihuku, a data analytics consultancy. Built an LLM document-automation system cutting manual creation time by 85% across three compliance regions, plus ML infrastructure and SQL/Python analytics serving multiple client engagements.

- Built an end-to-end document-generation system with large language models on Azure ML Studio, cutting manual creation time by 85% while meeting compliance requirements across three regions
- Integrated Twilio Programmable Messaging into a real-time customer notification system, achieving 95% timely delivery of alerts and updates
- Automated CI/CD pipelines and streamlined workflows, reducing deployment time by 40% and processing time by 35% across client engagements
- Built a face recognition system with Python and TensorFlow, improving processing efficiency by 25% through algorithm optimisation and model tuning
- Deployed ML models via FastAPI on Microsoft Azure, serving predictive analytics for agriculture (500+ farmers), financial inclusion (90% classification accuracy), and compliance systems
- Analysed large customer datasets in SQL and Python, delivering operational dashboards and optimising transaction workflows

EDUCATION

PhD · University College Cork

In progress

Neural engineering and machine learning

Doctoral research on segmenting EEG into microstate sequences using variational autoencoders, supervised by Luca Longo at University College Cork.

MSc · Technological University Dublin

Machine learning, deep learning, data mining, visualisation, statistics

Taught master's in computer science with a machine-learning concentration; coursework in deep learning, data mining, statistical modelling, and visualisation. Completed 2025 with a 4.0/4.0 GPA (80/100).

BSc · University of Eswatini

Information technology

Bachelor of Science in Information Technology, awarded with Second Class Honours, Upper Division (2.1), GPA 4.51.

TECHNICAL SKILLS

AI & MACHINE LEARNING	LLMs and generative AI (Azure OpenAI, OpenAI, Anthropic, DeepSeek, Llama) · Retrieval-augmented generation (RAG), LangChain, FAISS · Transformers and NLP · PyTorch · TensorFlow · scikit-learn · XGBoost / LightGBM / CatBoost · Anomaly and fraud detection · LLM fine-tuning (LoRA) · Explainable AI
PROGRAMMING	Python · SQL · Go · TypeScript · R · Bash
CLOUD & MLOPS	Azure (ML Studio, AI Studio, Functions, DevOps, Bicep) · AWS (SageMaker, Lambda, EC2, S3, Glue) · Docker · CI/CD (GitHub Actions, Azure DevOps, Jenkins) · Databricks
DATA ENGINEERING	ETL/ELT pipelines · Apache Spark · Apache Kafka · Airflow · Feature engineering · Vector databases
SYSTEMS & APIS	RESTful API design · Microservices and serverless · MySQL · PostgreSQL · Redis · Domain-driven design
ANALYTICS & STATISTICS	Statistical hypothesis testing (Wilcoxon, ICC, multiple-comparison correction) · Power BI · Tableau · Matplotlib · Seaborn

SELECTED WORK

- Founding engineer at Curnance. Seven production services on Azure Functions and MySQL: wallet and double-entry ledger, auth, tiered KYC/KYB, a Go jobs handler, a Flutter app, and an admin console. Payments through VFD, Flutterwave, and Paystack across eight African markets.
- Built the fraud/AML intelligence layer as rules first, LLM last. 38 deterministic detectors flag risk; the LLM only narrates anonymised aggregates, drafting suspicious-activity reports and answering compliance questions under k-anonymity floors and PII redaction.
- After a live withdrawal-fraud incident, rebuilt every external payout path to debit before paying, with row-level locking and deterministic idempotency keys. That race-condition class is gone.
- LLM document-generation pipeline on Azure ML Studio (Etihuku), cutting manual document work by 85% across three compliance regions.
- For Gijima, an Azure OpenAI (GPT-4) proposal-generation API with structured prompting, a deterministic pricing guardrail, SharePoint/Graph as the system of record, and a LangChain + FAISS contract-analysis module.
- Peer-reviewed research: Conv-VaDE for EEG microstate discovery (Brain Informatics 2026; N = 203, Wilcoxon signed-rank, BH correction) and a 4,832-model architecture search on SLURM GPU and IBM Power9 HPC (XAI 2026), released open-source under MIT.
- Now building an in-app banking assistant that turns natural-language requests (balance, exchange rate, recent transactions) into app actions.

Curnance, founding engineer: seven-service Azure fintech with a double-entry ledger and fraud/AML AI across eight African markets

Etihuku: LLM document automation on Azure ML Studio, 85% less manual work across three compliance regions

Gijima: GPT-4 proposal-generation API with deterministic pricing and RAG contract analysis

Conv-VaDE for EEG microstate discovery, Brain Informatics 2026 (doi:10.1186/s40708-026-00327-9)

4,832-model VAE architecture search, XAI 2026, open-source (MIT)

AI-assisted Farmer Call Center, UNESCO India-Africa Hackathon 2022 gold (AGRI12)

Swahili sentiment analysis (RoBERTa/BERT ensemble), 4th of 29, Google NLP Hack Series 2023

RECOGNITION

Google NLP Hack Series: Swahili Sentiment Analysis 2023

Google NLP Hack Series · 4th of 29

UmojaHack Africa 2023: Cryptojacking Detection 2023

UmojaHack Africa · 35th of 328 (top 11%)

Deep Learning IndabaX: Churn Prediction Challenge 2022

Deep Learning IndabaX · 3rd of 21

Winner, UNESCO India-Africa Hackathon 2022 (AGRI12) 2022

UNESCO · Ministry of Education Innovation Cell (India) · Gold medal + ₹3 lakh team prize

Tech4MentalHealth Challenge 2020

Basic Needs Basic Rights Kenya · 69th of 499 (top 14%)

SELECTED PROJECTS

Curnance

Multi-asset fintech platform for African markets. Seven production services spanning a double-entry ledger, tiered KYC/KYB, and a fraud/AML intelligence layer. · Founding engineer

`TypeScript · Go · Flutter · MySQL · Azure Functions · Azure DevOps · Bicep`

Etihuku document automation

End-to-end LLM document-generation system that cut manual creation time by 85% across three compliance regions. · Data Scientist

`Azure ML Studio · LLMs · Python · SQL`

AI-assisted Farmer Call Center

Automated voice-response system letting farmers report issues via phone, SMS, or web and receive AI-routed answers. · Team engineer, Eswatini representative

`Python · Voice AI · Twilio`

EEG microstate analysis with variational autoencoders

Source segmentation of EEG signals via variational autoencoders, including a GMM-VAE for soft clustering. · PhD researcher

`Python · PyTorch · MNE · NumPy · scikit-learn`

PUBLICATIONS

Morteza Akbari, Saheed Akinpelumi Faremi, Luca Longo "Autoencoder-Based Models for Scalp EEG: A Systematic Review of Architectures, Applications, Latent Representations, Interpretability, and Validation". *OSF Registries (registered systematic-review protocol),* 2026. doi:10.17605/osf.io/v2w3z

Saheed Faremi, Luca Longo "Integrating Convolutional Variational Autoencoders and the Gaussian Mixture Model for efficient manifold learning and clustering of spatially preserved EEG topographic maps". *Brain Informatics,* 2026. doi:10.1186/s40708-026-00327-9

Saheed Faremi, Andrea Visentin, Luca Longo "Interpretable EEG Microstate Discovery via Variational Deep Embedding: A Systematic Architecture Search with Multi-Quadrant Evaluation". *XAI 2026 (Late-breaking work + Doctoral Consortium track), Fortaleza, Brazil. arXiv preprint.,* 2026.

Saheed Faremi "Explainable Disentangled Representation Learning of Recurring Brain Activation Patterns via Variational Autoencoders". *XAI World Conference 2025, Doctoral Proposals track,* 2025.

Akinpelumi Saheed Faremi, Boluwaji Akinnuwesi, Elliot Mbunge, Petros M. Mashwama, Stephen Fashoto, Polite Zenzo Ncube, John Batani, Shamsudeen Ademola Sanni, Yinusa A. Faremi, Andile Metfula "Machine Learning Models for Identifying Factors Influencing and Predicting Malaria Among Children Under Five Years in Nigeria". *IEEE ICTAS 2024,* 2024. doi:10.1109/ICTAS59620.2024.10507142

TALKS

Interpretable EEG Microstate Discovery via Variational Deep Embedding · XAI 2026 (Late-breaking work + Doctoral Consortium track), 2026

Machine Learning Models for Predicting Malaria in Nigerian Children Under Five · IEEE ICTAS 2024, 2024